



# SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

## Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

## Case Study No.: 4

### 1. Case Study Title: ATM Card Pin Verification using JavaScript Functions Case Study.

### 2. Case Study Program Code:

```
// Global Scope: System configurations
//Name: Sonal Nakhate PRN: 24070521199
const BANK_NAME = "Global Secure Bank";
let dailyAttemptsLimit = 3;

// 1. FUNCTION DECLARATION: Reverses a PIN string or number
function reversePin(pin) {
  const pinString = String(pin);
  const reversed = pinString.split("").reverse().join("");
  return reversed;
}

// 2. FUNCTION EXPRESSION: Checks if a PIN is a palindrome
const isPalindromePin = function(pin) {
  const pinString = String(pin);
  const reversedPin = reversePin(pinString);
  return pinString === reversedPin;
};

// 3. ARROW FUNCTION: Formats output messages based on verification
const generateSecurityAlert = (pin, isPalindrome) => {
  if (isPalindrome) {
    return `[SPECIAL SECURITY PROTOCOL]: PIN ${pin} is a Palindrome.
High-symmetry alert triggered. Proceeding to main menu...`;
  }
}
```



# SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

## Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

```
    return `[STANDARD SECURITY]: PIN ${pin} verified successfully.
    Proceeding to main menu...`;
};

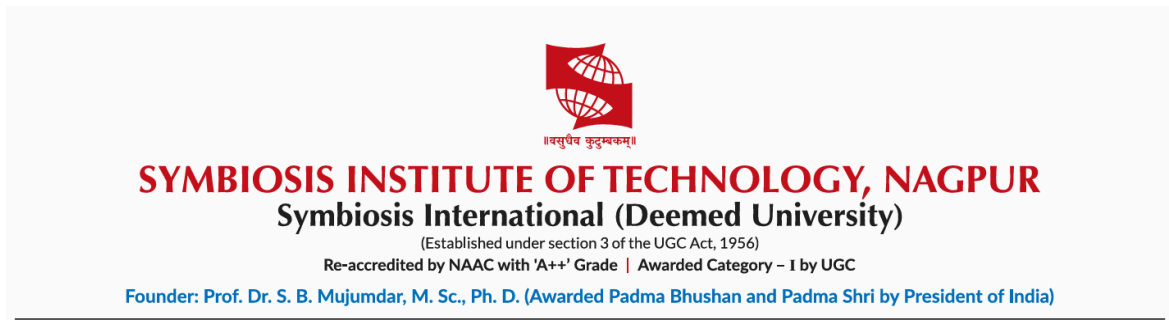
// 4. SCOPE DEMONSTRATION: Verification Handler
function processATMTransaction(enteredPin) {
    let attemptCounter = 1;

    console.log(`--- Processing PIN verification at ${BANK_NAME} ---`);

    if (attemptCounter <= dailyAttemptsLimit) {
        const isPalindromic = isPalindromePin(enteredPin);
        const securityMessage = generateSecurityAlert(enteredPin,
isPalindromic);
        console.log(securityMessage);

        // FIXED OBJECT RETURN: Comma added after true
        return {
            success: true,
            isPalindrome: isPalindromic
        };
    } else {
        console.log("Account locked. Exceeded daily limits.");
        return {
            success: false,
            isPalindrome: false
        };
    }
}

// Execution
processATMTransaction(1221);
processATMTransaction(4821);
```



### 3. Output:

The screenshot shows a Visual Studio Code editor with a JavaScript file named 'CS-ATMpin.js'. The code defines a 'reversePin' function and an 'isPalindromePin' function. The 'isPalindromePin' function uses 'reversePin' to check if a PIN is a palindrome. The terminal output shows the execution of the program, which processes PIN verification at 'Global Secure Bank'. It shows two test cases: 'PIN 1221 is a Palindrome. High-symmetry alert triggered. Proceeding to main menu...' and 'PIN 4821 verified successfully. Proceeding to main menu...'. The status bar at the bottom indicates 'Ln 11, Col 2' and 'Spaces: 4 UTF-8 CRLF {} JavaScript'.

**4. Result/Conclusion:** We have successfully implemented a simple JavaScript program that checks if the ATM pin is a palindrom or not.